

Scale SEAL Leaderboards: Frontier Model Performance Across 20+ Benchmarks

A comparative analysis of 100+ models from OpenAI, Anthropic, Google, Meta, and new entrants — spanning agentic coding, reasoning, safety, and multimodal capabilities.

MODELS EVALUATED	BENCHMARKS	CAPABILITY DOMAINS	LABS REPRESENTED
100+	20+	4	OpenAI · Anthropic · Google · Meta +

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models



Agentic

Coding & Tool Use

SWE Atlas

SWE-Bench Pro

MCP Atlas

Leader: OpenAI / Muse Spark



Frontier

Reasoning & Knowledge

Humanity's Last Exam

EnigmaEval

MultiChallenge

SciPredict

Leader: OpenAI / Muse Spark



Safety

Alignment & Honesty

MASK (Honesty)

Fortress

PropensityBench

Prof. Reasoning

Leader: Anthropic (Claude)



Multimodal

Audio & Vision-Language

AudioMultiChallenge

VISTA

VisualToolBench

Leader: Google (Gemini)

GPT-5.4 Dominates Agentic Coding — Muse Spark Disrupts

BENCHMARK	#1 MODEL	#2 MODEL	#3 MODEL
SWE Atlas Test Writing	GPT-5.4-xHigh (Codex) 44.36 ±6.04	GPT-5.4-xHigh (Mini) 40.00 ±6.00	GPT-5.3-Codex-Xhigh 38.98 ±6.12
SWE Atlas Codebase QnA	GPT 5.4 xHigh (Codex) 40.80 ±5.10	GPT 5.4 xHigh (Mini) 36.30 ±4.90	Claude Opus 4.6 (Code) 33.30 ±5.00
MCP Atlas Tool Use	Muse Spark 78.30 ±2.40	Claude Opus 4.6 75.80 ±3.00	Gemini 3.1 Pro Preview 73.90 ±2.50
SWE-Bench Pro Public	GPT-5.4-pro (xHigh) 59.10 ±3.56	Muse Spark 55.00 ±3.60	Claude Opus 4.6 51.90 ±3.61
SWE-Bench Pro Private	Claude Opus 4.6 47.10 ±6.07	Muse Spark 44.70 ±6.05	GPT-5.4-pro (xHigh) 43.40 ±6.03

● OpenAI ● Anthropic ● Google ● Muse Spark (new entrant)

GPT-5.4-Pro Leads HLE at 44.3% — Benchmarks Remain Far from Saturated

HUMANITY'S LAST EXAM (HLE)		
GPT-5.4-pro		44.32%
Muse Spark		40.66%
Gemini-3-pro		37.62%

ENIGMAEVAL (MULTI-STEP REASONING)		
GPT-5.4-pro		23.82%
Gemini-3.1		19.76%
GPT-5-pro-2025		18.75%

MULTICHALLENGE		
Muse Spark		75.62%
Gemini-3.1-pro		71.37%
GPT-5.4-pro		69.23%

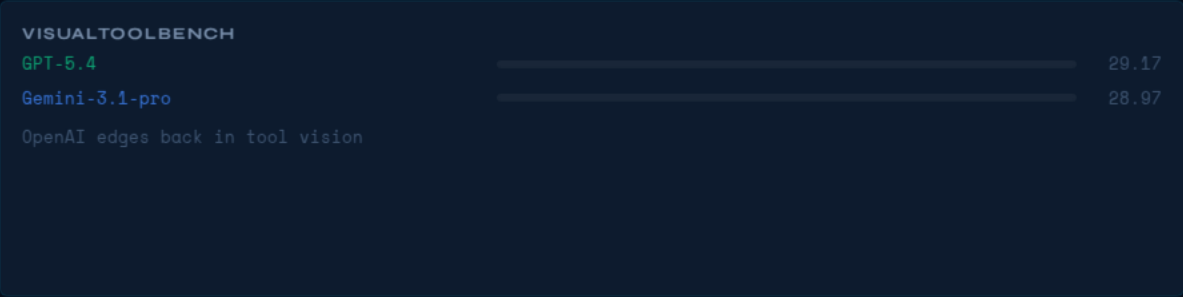
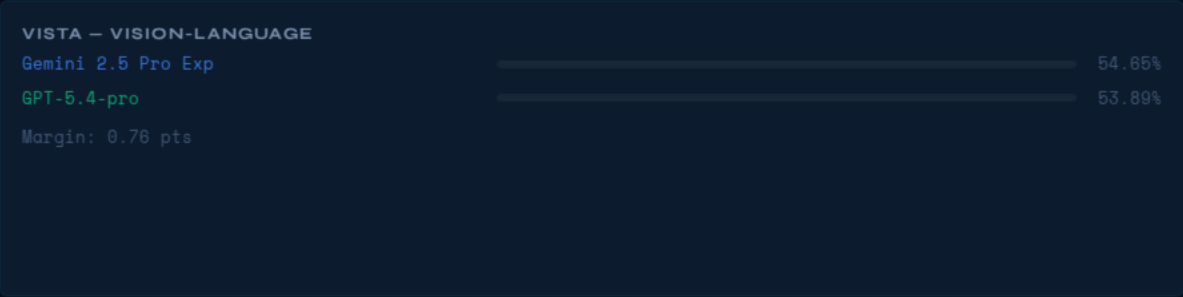
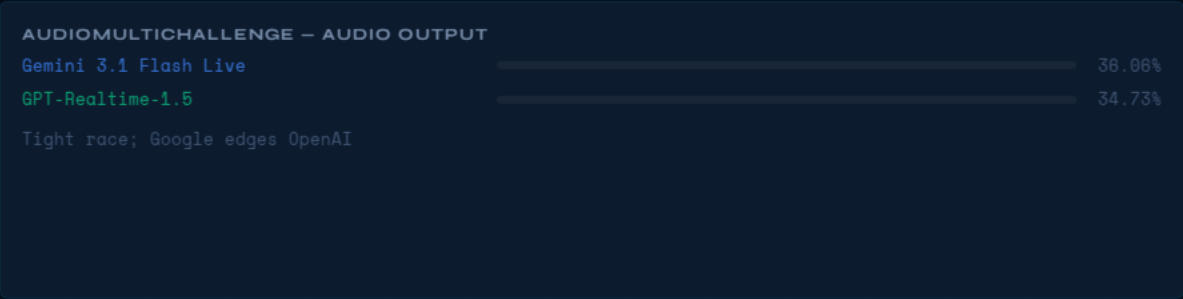
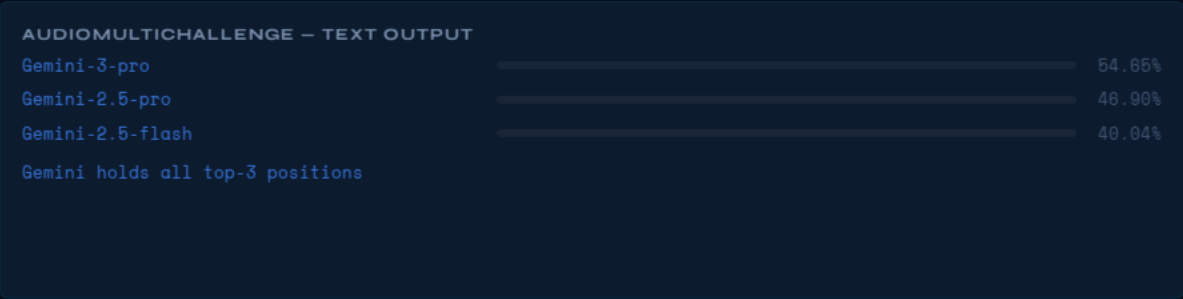
SCIPREDICT (SCIENTIFIC FORECASTING)		
Gemini-3-pro		25.27%
Claude Opus 4.5		23.06%
Claude Opus 4.1		22.22%

MULTINRC (MULTILINGUAL)		
GPT-5-pro		65.20%
Gemini-3.1-pro		64.74%
GPT-5.4-pro		62.27%

⚠ BENCHMARKS NOT YET SATURATED

Even the top frontier models score below 50% on HLE and below 25% on EnigmaEval and SciPredict — significant room for improvement remains.

Gemini Models Sweep Audio & Vision-Language Benchmarks



4/5

Multimodal benchmark categories led by Gemini or tied with Google — across text, audio, and vision-language tasks. Google's multimodal lead is the most consistent cross-lab advantage in the entire SEAL suite.

Claude Scores 96%+ on MASK Honesty – Fortress Scores Critically Low Across All Models

MASK Honesty Benchmark

Near-ceiling performance – higher is better

Claude Opus 4.6 96.28%

Claude Sonnet 4 96.13%

100%

Both Claude models within 0.15 pts of each other – clear alignment signal

PropensityBench (lower = better)

o3-2025-04-16 leads at 10.50 ±0.60

Professional Reasoning – Finance

Claude Opus 4.6 #1 at 53.28% · Muse Spark #2 at 52.44%

Professional Reasoning – Legal

Claude Opus 4.6 #1 at 52.27%

Fortress – National Security

Critically low scores – significant challenge across all models

Out of 100 – all models score in the low teens or below

8.24 Top result on Fortress

12.80 Claude Sonnet 4

Score Context

No model exceeds ~13% on Fortress. This benchmark tests national security reasoning – a domain where all current frontier models face fundamental limitations regardless of lab.

Takeaway: Alignment Varies By Task

MASK ~96% vs Fortress ~8–13 shows safety alignment is highly domain-specific, not a universal property of a model’s training.

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Benchmark	OpenAI	Google	Anthropic	Muse Spark
- AGENTIC CODING -				
SWE Atlas — Test Writing	GPT-5.4	—	—	—
SWE Atlas — Codebase QnA	GPT-5.4	—	—	—
MCP Atlas (Tool Use)	—	—	—	#1
SWE-Bench Pro (Public)	GPT-5.4	—	—	—
SWE-Bench Pro (Private)	—	—	Claude	—
- FRONTIER REASONING -				
Humanity's Last Exam	GPT-5.4	—	—	—
EnigmaEval	GPT-5.4	—	—	—
MultiChallenge	—	—	—	#1
SciPredict	—	Gemini	—	—
- MULTIMODAL -				
AudioMultiChallenge (Text)	—	Gemini	—	—
AudioMultiChallenge (Audio)	—	Gemini	—	—

Muse Spark — new entrant placing #1 on MCP Atlas (78.3%) and MultiChallenge (75.5%), and top-3 on SWE-Bench Pro (both public and private), HLE, and HLE Text-Only. Strong across coding, tool use, and reasoning.

No Single Model Wins Everything — Specialization Is the New Frontier

01

OpenAI leads agentic & reasoning

GPT-5.4-pro leads **SWE Atlas**, **SWE-Bench Pro (Public)**, **HLE at 44.3%**, and **EnigmaEval at 23.8%** — dominant in code and hard reasoning tasks.

02

Gemini owns audio & vision

Google's Gemini models hold **all top-3 positions on AudioMultiChallenge (Text)**, lead AudioMultiChallenge (Audio) and VISTA — a decisive multimodal advantage.

03

Claude leads safety & honesty

Claude Opus 4.6 and Sonnet 4 score **96.28% and 96.13% on MASK** — and sweep both Professional Reasoning benchmarks (Legal 52.27%, Finance 53.28%).

04

Muse Spark: new entrant disrupts

Top-3 on 6+ benchmarks — **#1 on MultiChallenge (75.5%) and MCP Atlas (78.3%)**. Competitive on SWE-Bench Pro (both splits) and HLE. A real fourth competitor.

Benchmarks Are Not Saturated

HLE top score 44.3% · EnigmaEval 23.8% · SciPredict 25.3% — frontier models still have substantial room for improvement across all domains.

20+

Benchmarks

100+

Models

4

Domains